

WORKED EXAMPLE

A slope tells us which way changes the loss

Use two parameters x and y , collected into the vector $w = (x, y)$.

$$E(x, y) = 1.5x^2 + xy + 1.5y^2$$

A partial derivative measures change when one parameter moves and the other stays fixed. The gradient collects these slopes into a vector.

$$\nabla E = \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y} \right)$$

Curvature means how a slope itself changes as we move. The Hessian collects derivatives of gradient components. Our tuples list those components; the notes write the gradient as a column vector.

This original study example is N4.3. It has the quadratic form on Lecture4 PDF page37 with A equal to its Hessian, and $b = 0$, $c = 0$.

WORKED EXAMPLE

Differentiate one parameter at a time

Treat the other parameter as a constant.

$$\frac{\partial E}{\partial x} = 1.5(2x) + y + 0 = 3x + y$$

For the xy term, y behaves like a fixed coefficient when x changes. The y -squared term is constant with respect to x .

$$\frac{\partial E}{\partial y} = 0 + x + 1.5(2y) = x + 3y$$

$$\nabla E(2, 0) = (3(2) + 0, 2 + 3(0)) = (6, 2)$$

At $(2,0)$, increasing x a little increases E about 6 times that increment; increasing y a little increases E about 2 times that increment.

WORKED EXAMPLE

Check what those slopes mean numerically

Small changes connect a derivative to something you can calculate.

$$E(2, 0) = 1.5(4) + 0 + 0 = 6$$

$$E(2.01, 0) = 1.5(2.01)^2 = 6.06015$$

$$\Delta E = 0.06015 \approx 6(0.01) = 0.06$$

$$E(2, 0.01) = 6 + 2(0.01) + 1.5(0.01)^2 = 6.02015$$

$$\Delta E = 0.02015 \approx 2(0.01) = 0.02$$

The slight difference is the quadratic curvature contribution. A derivative is a local rate, not an exact finite-change formula.

WORKED EXAMPLE

Differentiate the slopes to form the Hessian

Rows name the gradient component; columns name the parameter being changed.

	differentiate by x	differentiate by y
gradient x: $3x+y$	$H_{xx} = 3$	$H_{xy} = 1$
gradient y: $x+3y$	$H_{yx} = 1$	$H_{yy} = 3$

Thus H has rows (3,1) and (1,3). The off-diagonal 1 values come from the mixed term xy : moving y changes the x -slope, and moving x changes the y -slope.

$$\Delta x = 0.01 \implies \Delta(3x + y) = 3(0.01) = 0.03$$

$$\Delta y = 0.01 \implies \Delta(x + 3y) = 1(0.01) = 0.01$$

These slope changes are exact here because the gradient is linear. The Hessian is constant for this quadratic.

WORKED EXAMPLE

Find a stationary point before classifying it

Stationary means every first derivative is zero.

$$3x + y = 0, \quad x + 3y = 0$$

$$y = -3x \implies x + 3(-3x) = -8x = 0$$

$$x = 0, \quad y = -3(0) = 0$$

A zero gradient says the first-order slope vanishes. It does not tell us whether the point is a minimum, maximum or saddle.

To distinguish them, inspect how the function curves in different directions near that point.

WORKED EXAMPLE

Eigenvalues describe directional curvature

An eigenvector is a nonzero direction v for which $H v = \lambda v$.

The matrix changes the length or sign of this direction, without rotating it to another line. λ is the corresponding scale factor. For a unit direction, it is the quadratic curvature.

I is the identity matrix: diagonal entries 1 and off-diagonal entries 0. $H - \lambda I$ therefore has rows $(3-\lambda, 1)$ and $(1, 3-\lambda)$.

For a 2-by-2 matrix with rows (a,b) and (c,d) , its determinant is $ad-bc$. Set this determinant to zero to find the possible eigenvalues.

$$\det(H - \lambda I) = (3 - \lambda)(3 - \lambda) - 1(1) = 0$$

$$(3 - \lambda)^2 = 1 \implies \lambda = 2 \text{ or } 4$$

WORKED EXAMPLE

Verify both eigenvector directions by multiplication

These two independent directions account for the whole two-parameter space.

$$\begin{aligned} H(1, -1) &= (3(1) + 1(-1), 1(1) + 3(-1)) \\ &= (2, -2) = 2(1, -1) \end{aligned}$$

$$\begin{aligned} H(1, 1) &= (3(1) + 1(1), 1(1) + 3(1)) \\ &= (4, 4) = 4(1, 1) \end{aligned}$$

Both eigenvalues are positive. This is called positive definite: every nonzero direction has positive quadratic curvature. At our stationary point, that establishes a strict local minimum. Local compares nearby points; strict means every different nearby point is higher. Global compares the entire domain.

WORKED EXAMPLE

Derive the two directional curves

Substitute the coordinates of each unit direction into the original function.

$$x = t/\sqrt{2}, \quad y = -t/\sqrt{2}$$

$$E = 1.5(t^2/2) - t^2/2 + 1.5(t^2/2) = t^2$$

$$x = t/\sqrt{2}, \quad y = t/\sqrt{2}$$

$$E = 1.5(t^2/2) + t^2/2 + 1.5(t^2/2) = 2t^2$$

$$\frac{d^2(t^2)}{dt^2} = 2, \quad \frac{d^2(2t^2)}{dt^2} = 4$$

The sign of xy changes between the directions; the two square terms stay the same. This is why the same surface has different curvature along different directions.

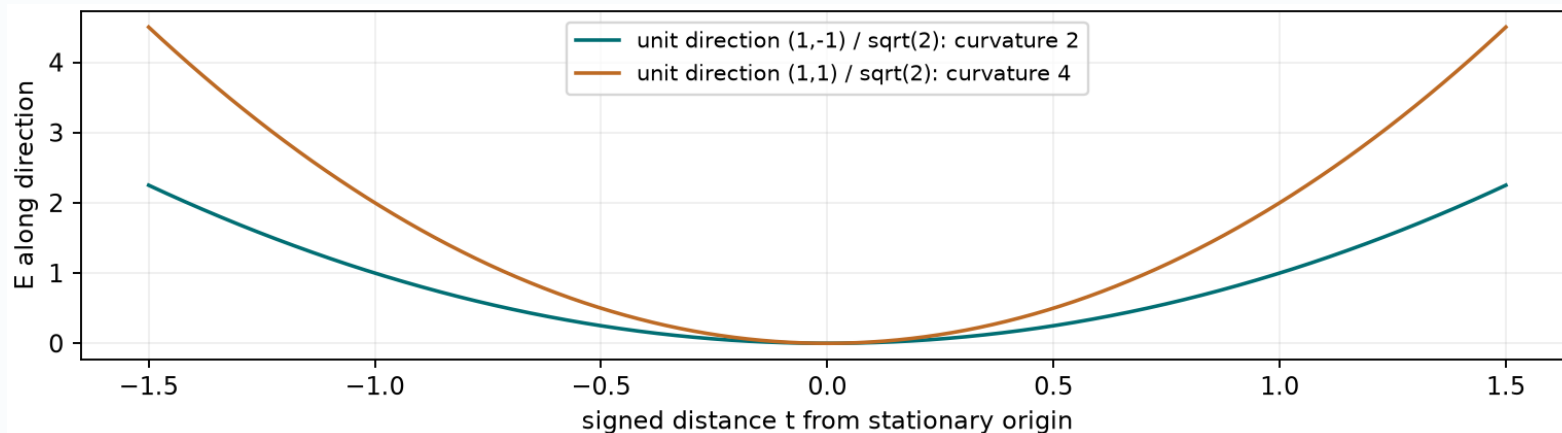
WORKED EXAMPLE

Normalize directions before comparing curvature

A unit direction has length 1, so the movement coordinate measures distance.

$$u_- = (1, -1)/\sqrt{2}, \quad u_+ = (1, 1)/\sqrt{2}$$

$$E(tu_-) = t^2, \quad E(tu_+) = 2t^2$$



The second derivatives with respect to distance t are 2 and 4, exactly the eigenvalues. Using unnormalized $(1,1)$ would give $E=4t$ -squared and second derivative 8 because each unit of t moves farther.

WORKED EXAMPLE

Maxima curve down; saddles mix directions

Evaluate nearby points to see the distinction directly.

$$M(x, y) = -E(x, y), \quad \nabla M(0, 0) = 0$$

Its Hessian is $-H$ with eigenvalues -2 and -4 . Both are negative, so $(0,0)$ is a strict maximum of M .

$$S(x, y) = \frac{x^2 - y^2}{2}, \quad \nabla S = (x, -y)$$

At $(0,0)$, the gradient is zero and the Hessian is diagonal $(1, -1)$. One curvature is positive, the other negative: the Hessian is indefinite.

$$S(0, 0) = 0, \quad S(0.1, 0) = 0.005, \quad S(0, 0.1) = -0.005$$

Nearby values occur above and below zero, so the origin is a saddle, not a minimum or maximum.

WORKED EXAMPLE

A zero Hessian can leave the test inconclusive

Use the original function when second-order information is insufficient.

$$T(x, y) = x^4 + y^4, \quad \nabla T = (4x^3, 4y^3)$$

The Hessian has diagonal entries $12x^2$ and $12y^2$, with zero off-diagonal entries. At $(0,0)$, every Hessian entry is zero.

$$\nabla T(0, 0) = (0, 0), \quad \lambda_1 = \lambda_2 = 0$$

The second-order test is inconclusive. It does not say there is no minimum.

$$T(x, y) \geq 0 = T(0, 0)$$

Fourth powers are nonnegative and both vanish only at the origin. This direct argument proves a strict global minimum.

INDEPENDENT PRACTICE

Your turn: classify two fresh functions

Find gradients, Hessians, stationary points and curvature signs.

$$Q(x, y) = 2x^2 + xy + 2y^2, \quad R(x, y) = -x^2 - 2y^2$$

For Q, also calculate the gradient at (1,-1), solve the two-by-two eigenvalue equation, and verify one eigenvector direction.

For R, explain why the stationary point is a maximum even though its gradient is zero, just like the minimum of Q.

State whether zero Hessian eigenvalues always mean a flat function. Use T from the preceding page to support your answer.

WORKED ANSWER

Practice Q: derivatives and stationary point

Differentiate each term while holding the other variable fixed.

$$Q_x = 2(2x) + y = 4x + y, \quad Q_y = x + 2(2y) = x + 4y$$

$$\nabla Q(1, -1) = (4(1) - 1, 1 + 4(-1)) = (3, -3)$$

Differentiate these slopes: H has rows (4,1) and (1,4). The mixed term contributes 1 to each off-diagonal entry.

$$4x + y = 0 \Rightarrow y = -4x$$

$$x + 4(-4x) = -15x = 0 \Rightarrow (x, y) = (0, 0)$$

WORKED ANSWER

Practice Q: eigenvalues and classification

Use the determinant and then check an actual direction.

$$\det(H - \lambda I) = (4 - \lambda)^2 - 1 = 0$$

$$4 - \lambda = \pm 1 \implies \lambda = 3, 5$$

$$H(1, -1) = (4 - 1, 1 - 4) = (3, -3) = 3(1, -1)$$

Both eigenvalues are positive. The Hessian is positive definite, so the stationary origin is a strict local minimum.

$$Q(x, y) = 1.5(x^2 + y^2) + 0.5(x + y)^2 \geq 0$$

The last expression also proves it is the global minimum: all terms are nonnegative and vanish together only at the origin.

WORKED ANSWER

Practice R: downward curvature and common errors

Stationarity and classification are separate questions.

$$\nabla R = (-2x, -4y) = (0, 0) \Rightarrow (x, y) = (0, 0)$$

Its Hessian is diagonal $(-2, -4)$, so its eigenvalues are -2 and -4 . Both are negative: the origin is a strict maximum.

$$R(x, y) = -x^2 - 2y^2 \leq 0 = R(0, 0)$$

Zero eigenvalues do not imply the function is constant or that no minimum exists.

$T = x^4 + y^4$ has zero curvature exactly at its minimum but grows away from it.

Common errors: treating the other variable as zero instead of constant, omitting mixed derivatives, calling every zero-gradient point a minimum, or using an unnormalized direction as distance.